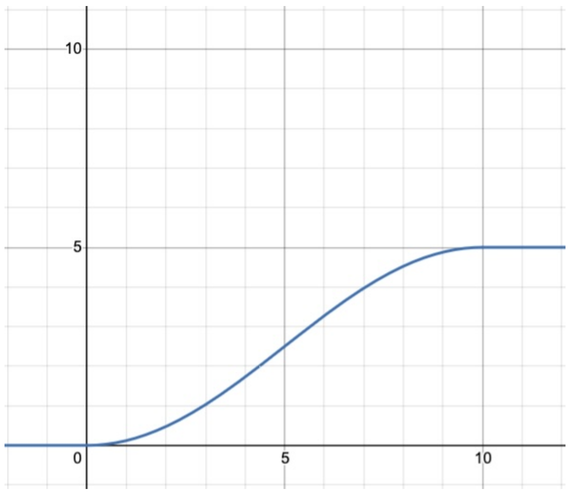
Learning target DF0, version 1

Compute and interpret average velocity and its units, using information about the position function (as a table, graph, and/or function formula).

The position (in miles), $s(t)$, of a car driving along a straight road at time t (in minutes), is given by the following graph.



1. Determine the average velocity of the car between $t = 5$ and $t = 10$ minutes, $AV_{[5,10]}$. Use proper notation and the work you did to determine the result; include units on your answer.

2. On the graph of $s(t)$ draw a line through $(3, s(3))$ and $(7, s(7))$; what is the slope of this line and what does that slope mean in the physical context of the function s?

3. Here's some additional data for the function $s(t)$ that's pictured above:

t (in minutes)	8.0	8.05	8.1	8.15
$s(t)$ (in miles)	4.52254	4.54537	4.56770	4.58952

   Find the average velocity of the car on the interval $[8.05, 8.1]$. Label your result using proper notation and include units on your answer.

MATH 201 Your name: ______________________________

Learning target DF0, version 2

Compute and interpret average velocity and its units, using information about the position function (as a table, graph, and/or function formula).

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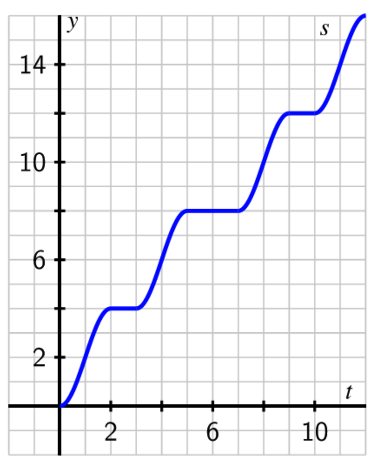
1. Your fitbit records some data about the distance you traveled on a bike ride around the city:

t (in minutes)	0	15	30	45	60	75
$d(t)$ (in miles)	0	4.5	10.5	15	21	24

Find your average velocity over the time interval from $t = 30$ to $t = 75$. Include units.

2. The following graph shows a plot of the function $s(t)$ that measures the location of a car (in thousands of feet) along a road at time t in minutes.



- Use the graph to find the average velocity of the car between $t = 2$ and $t = 7$ minutes, $AV_{[2,7]}$. Include units on your answer.

- On the graph of $s(t)$ draw a line through $(7, s(7))$ and $(10, s(10))$; what is the slope of this line, and what does that slope mean in the physical context of the function s? Include units on your answer.

MATH 201 Your name: ____________________

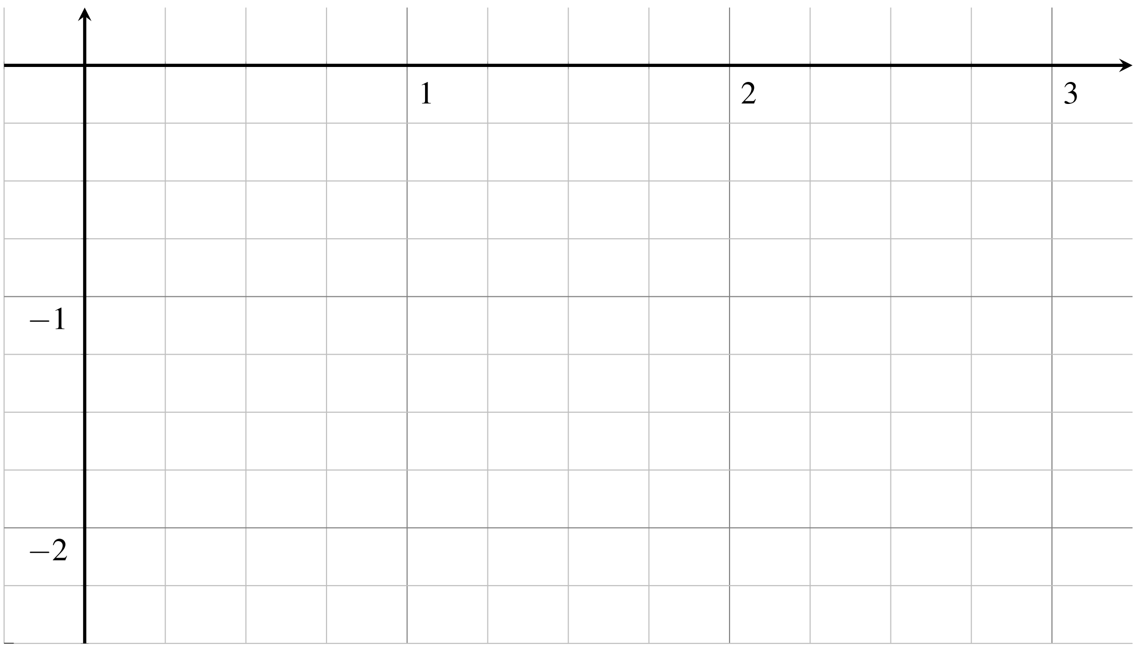
Learning target DF1, version 1

Estimate the value of a derivative using difference quotients, and draw corresponding secant and tangent lines on the graph of a function.

Suppose that you know the following values of some function $g(x)$:

x	1.7	2	2.3
$g(x)$	-2.2	-2	-1.5

1. Plot these points and sketch a plausible graph of $g(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $g(x)$ at $x = 2$.
3. Use a central difference to estimate $g'(2)$. Draw the corresponding secant line on your graph above.

4. Compute another, different estimate of $g'(2)$. Draw the corresponding secant line on your graph.

5. Which one of your approximations is the best? How do you know?

MATH 201 Your name: ____________________

Learning targets DF1 and DFa, version 2

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table[1] gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t.

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.

2. Write a sentence explaining what the number means about how the area of the glacier is changing. **Don't say "per," and don't say "rate."**

3. Compute an approximation for $A'(1999)$, and **include units** for this number.
   Do you think your approximation is too high or too low? Why?

4. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

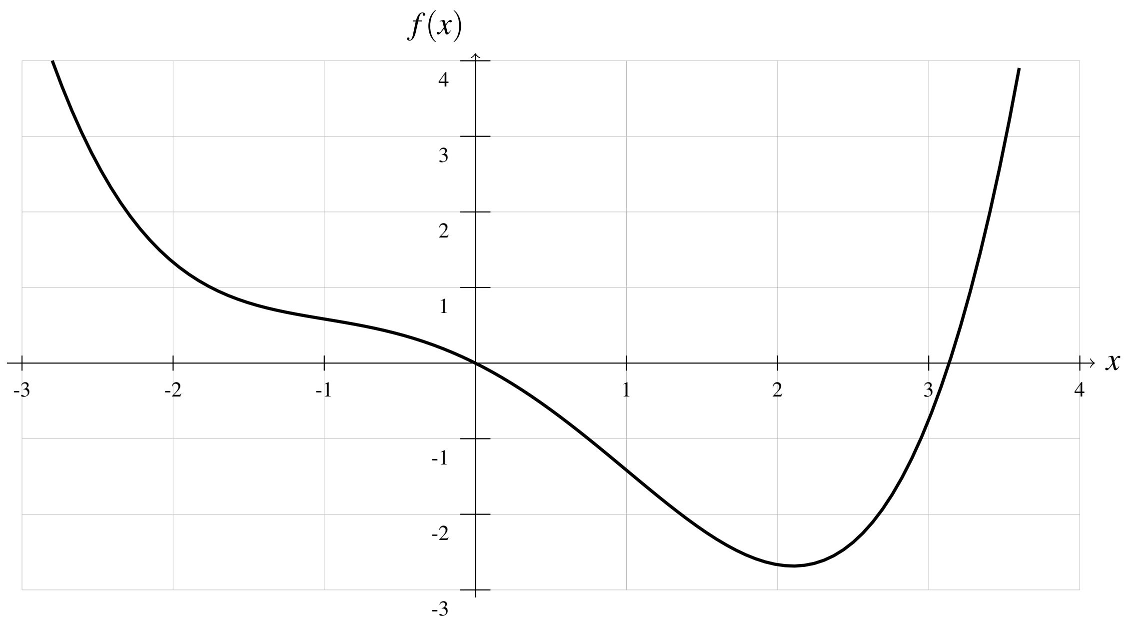
[1] Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research, 42*(2), 198-209.

MATH 201 Your name: ____________________

Learning target DF1, version 3

Estimate the value of a derivative using difference quotients, and draw corresponding secant and tangent lines on the graph of a function.

Below is a graph of the function $f(x) = \dfrac{x^4}{12} - \dfrac{x^2}{2} - x$.



(a) On the graph above, draw the tangent line to the curve at $x = -1$.

(b) On the graph above, draw a secant line that goes through the curve at $x = -1$ and at some other nearby x-value of your choice.

(c) Calculate the slope of that secant line.

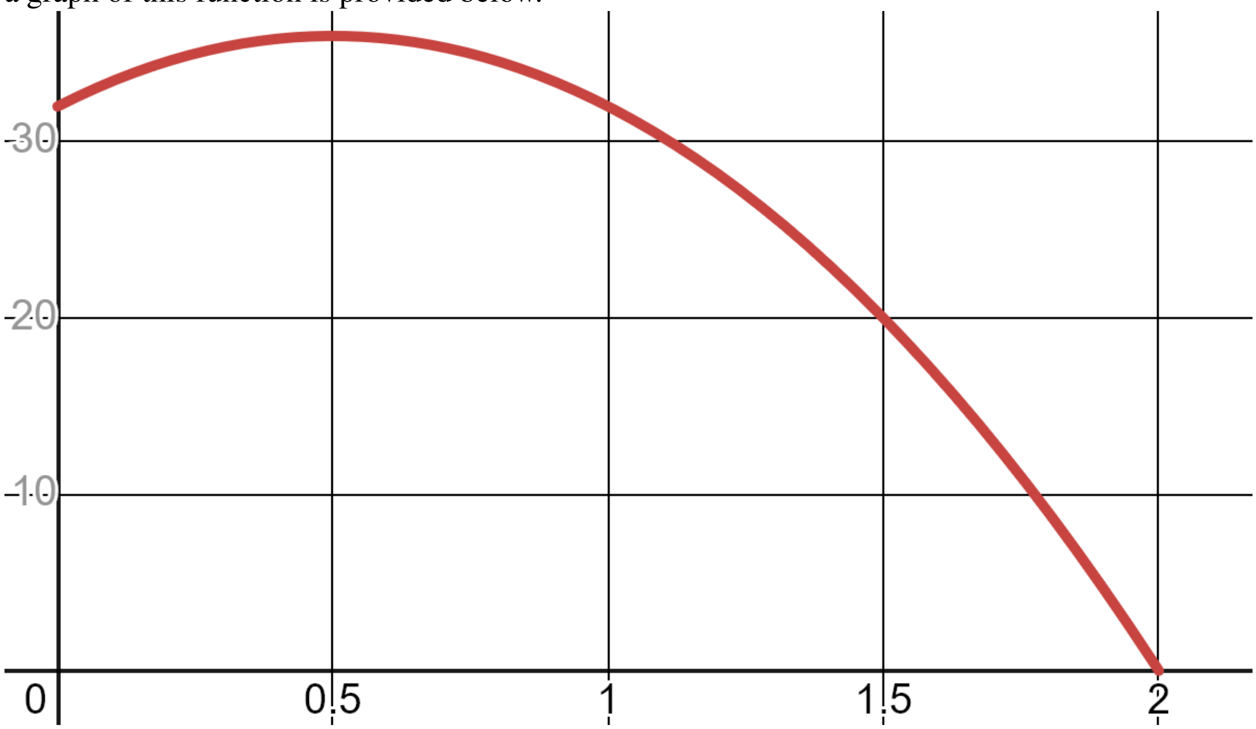
(d) Use $f'(x)$ to calculate the slope of the tangent line. Compare the value you get here to the value you got in part (c). Does that make sense?

MATH 201 Your name: ______________________________

Learning target DF1, version 4

Estimate the value of a derivative using difference quotients, and draw corresponding secant and tangent lines on the graph of a function.

A water balloon is tossed vertically in the air from a window. The balloon's height (measured in feet) at time t (measured in seconds after being launched) is given by $s(t) = -16t^2 + 16t + 32$; a graph of this function is provided below.



(a) Compute $s(1)$. Show your work.

(b) On the graph above, carefully sketch the *tangent* line to $s(t)$ at $t = 1$.

(c) On the graph above, carefully sketch a *secant* line through the point on the graph at $t = 1$ and a second nearby point.

(d) Compute the slope of your secant line. Show your work and **include units**.
Hints: rise over run; one of your two y-values is $s(1)$, which you computed in part (a).

(e) Use shortcut rules to find $s'(t)$, and compute $s'(1)$; show your work. Compare your result here to your result in part (d). Do these two numbers make sense together? Why?

MATH 201 Your name: ______________________

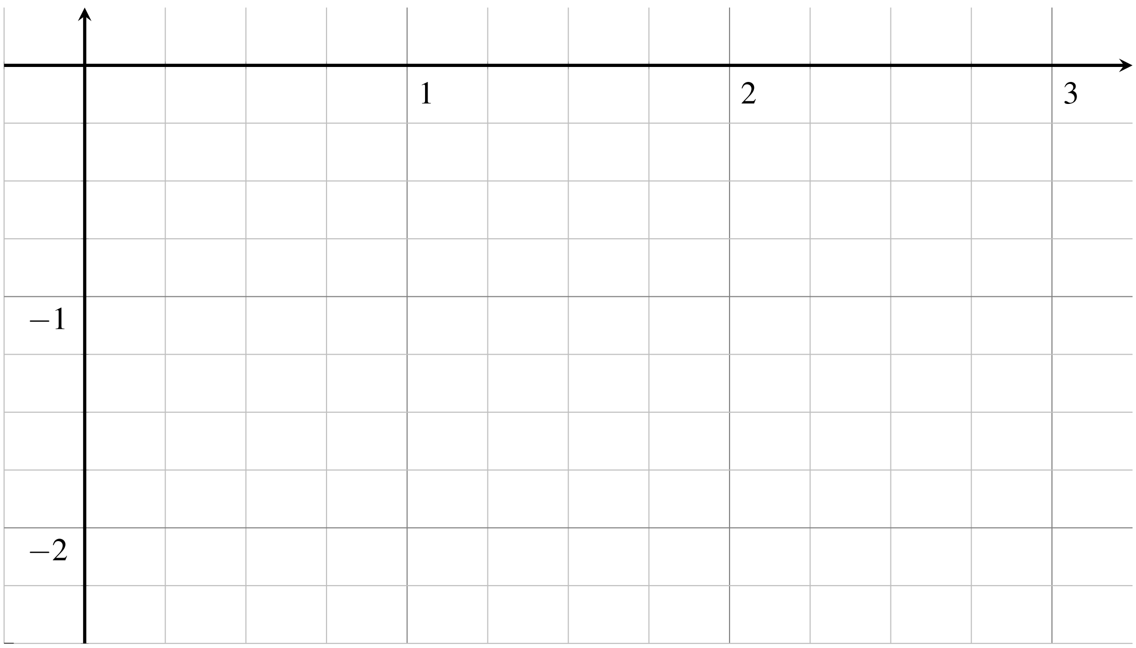
Learning target DF1, version 5

Estimate the value of a derivative using difference quotients, and draw corresponding secant and tangent lines on the graph of a function.

Suppose that you know the following values of some function $f(x)$:

x	1	1.5	2
$f(x)$	-0.8	-1	-1.6

1. Plot these points and sketch a plausible graph of $f(x)$.



2. On your graph above, draw a plausible tangent line to the graph of $f(x)$ at $x = 1.5$.
3. Use a central difference to estimate $f'(1.5)$. Draw the corresponding secant line on your graph.

4. Compute another, different estimate of $f'(1.5)$. Draw the corresponding secant line on your graph.

5. Which one of your approximations is the best? How do you know?

MATH 201 Your name: ____________________

Learning target DF2, version 1

Find derivatives using the definition of derivative as a limit.

Suppose that $f(x) = 3x^2 - 5x + 4$. Use the limit definition of the derivative to find $f'(x)$.

PS: The answer is $f'(x) = 6x - 5$.

MATH 201 Your name: ____________________

Learning target DF2, version 2

Find derivatives using the definition of derivative as a limit.

Suppose that $g(w) = 6w^3 - 2w^2 - 9w - 2$. Use the limit definition of the derivative to find $g'(w)$.

Algebra hint: $(w+h)^3 = w^3 + 3w^2h + 3wh^2 + h^3$.

PS: The answer is $g'(w) = 18w^2 - 4w + 9$.

MATH 201 Your name: ____________________

Learning target DF2, version 3

Find derivatives using the definition of derivative as a limit.

Suppose that $p(z) = -2z^3 + 5z^2 - 4z + 4$. Use the limit definition of the derivative to find $p'(z)$.

Algebra hint: $(z+h)^3 = z^3 + 3z^2h + 3zh^2 + h^3$.

PS: The answer is $p'(z) = -6z^2 + 10z - 4$.

MATH 201 Your name: ____________________

Learning target DF2, version 4

Find derivatives using the definition of derivative as a limit.

Suppose that $f(x) = 4x^2 - 3x + 100$. Use the limit definition of the derivative to find $f'(x)$.

PS: The answer is $f'(x) = 8x - 3$.

MATH 201 Your name: ____________________

Learning target DFa, version 1

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

A company manufactures rope, and the total cost of producing r feet of rope is $C(r)$ dollars.

1. Suppose $C(2000) = 800$. What are the units on the 2000? What are the units on the 800?

2. Write a sentence explaining what $C(2000) = 800$ means in the story.

3. Suppose $C'(2000) = 0.35$. What are the units on the 2000? What are the units on the 0.35?

4. Write a sentence explaining what $C'(2000) = 0.35$ means in the story.
   Don't say "per," and don't say "rate."

MATH 201 Your name: ____________________

Learning targets DF1 and DFa, version 2

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table[2] gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t.

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.

2. Write a sentence explaining what the number means about how the area of the glacier is changing. **Don't say "per," and don't say "rate."**

3. Compute an approximation for $A'(1999)$, and **include units** for this number.
Do you think your approximation is too high or too low? Why?

4. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

[2]Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research, 42*(2), 198-209.

MATH 201 Your name: ________________________

Learning target DFa, version 3

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

Let $f(t)$ be the number of centimeters of rain that have fallen by time t hours after midnight.

(a) Suppose $f(3) = 1.5$. What are the units on the 3? What are the units on the 1.5?

(b) Write a sentence explaining what $f(3) = 1.5$ means in the story.

(c) Suppose $f'(3) = 0.2$. What are the units on the 3? What are the units on the 0.2?

(d) Write a sentence explaining what $f'(3) = 0.2$ means in the story.
Don't say "per" and don't say "rate".

(e) Would the statement $f'(5) = -0.7$ make sense? Why or why not?

MATH 201 Your name: ____________________

Learning target DFa, version 4

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

In order to prepare for the upcoming ski season, a local resort is making snow on some of its lower trails. The snow depth, in inches, at time t is $D(t)$, where t is the number of days we are into December (so, for instance, $t = 9$ is December 9).

(a) Suppose $D(15) = 28$. What are the units on the 15? What are the units on the 28?

(b) Write a sentence explaining what $D(15) = 28$ means in the story.

(c) Suppose $D'(15) = 1.4$. What are the units on the 15? What are the units on the 1.4?

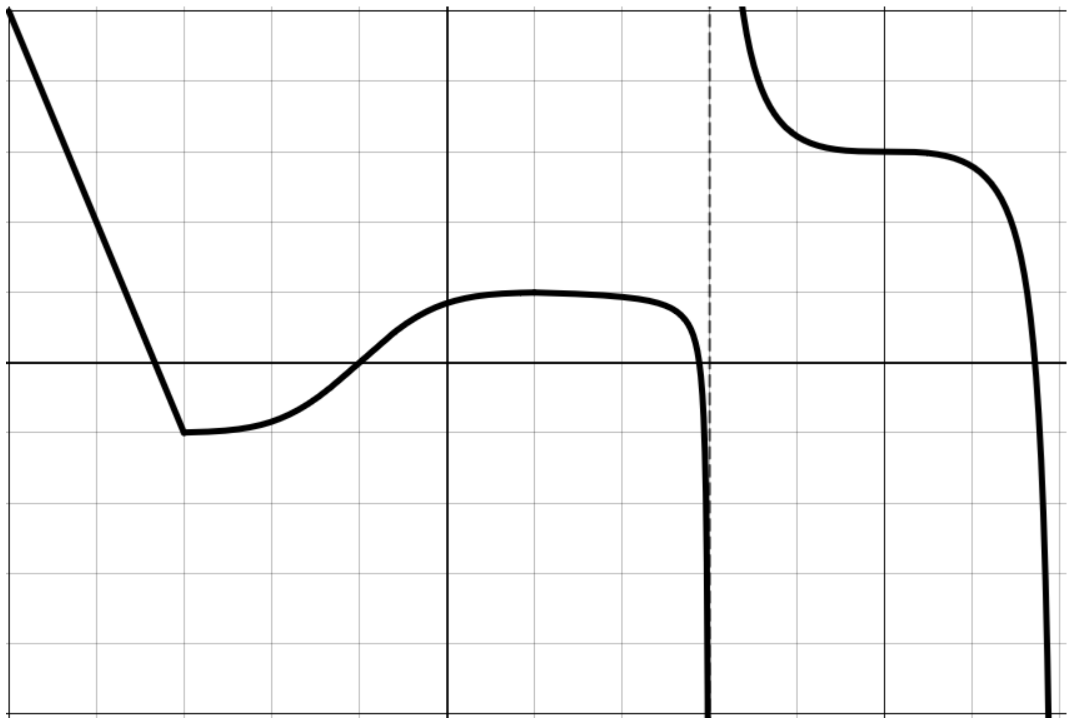
(d) Write a sentence explaining what $D'(15) = 1.4$ means in the story.
Don't say "per" and don't say "rate".

MATH 201 Your name: ____________________

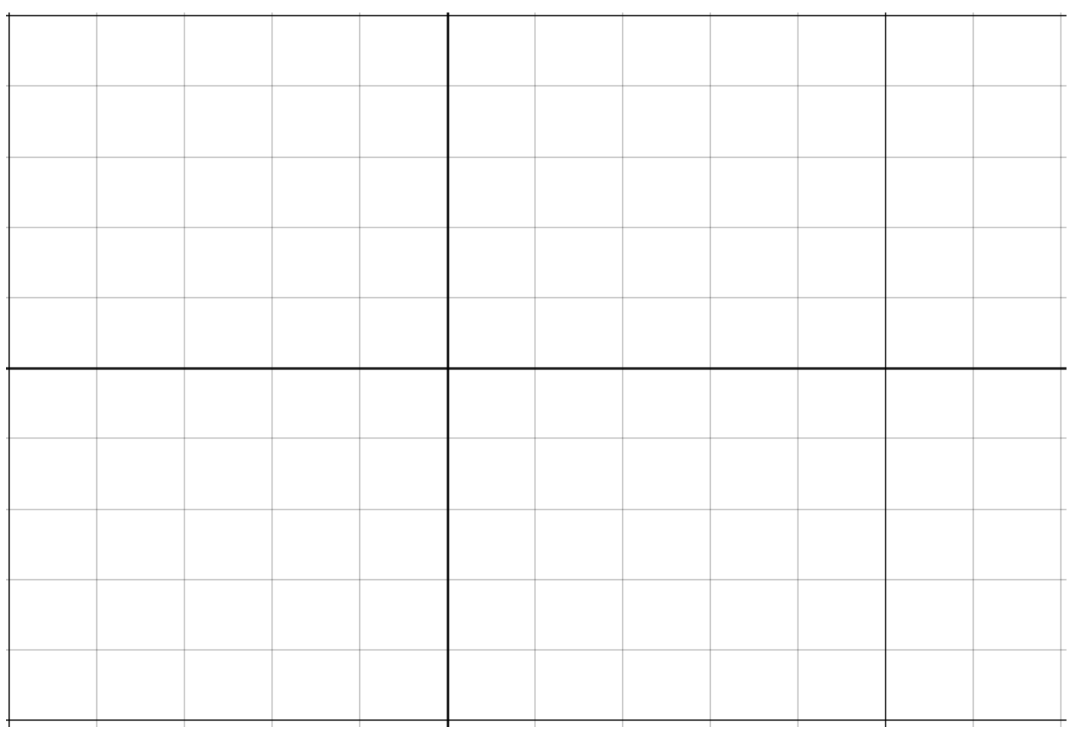
Learning target DFb, version 1

Given a graph of a function, estimate specific values of the derivative, and/or produce a sketch of the derivative function.

Here is the graph of some wacky function $q(x)$:



Sketch the graph of $q'(x)$ on the blank axes below.

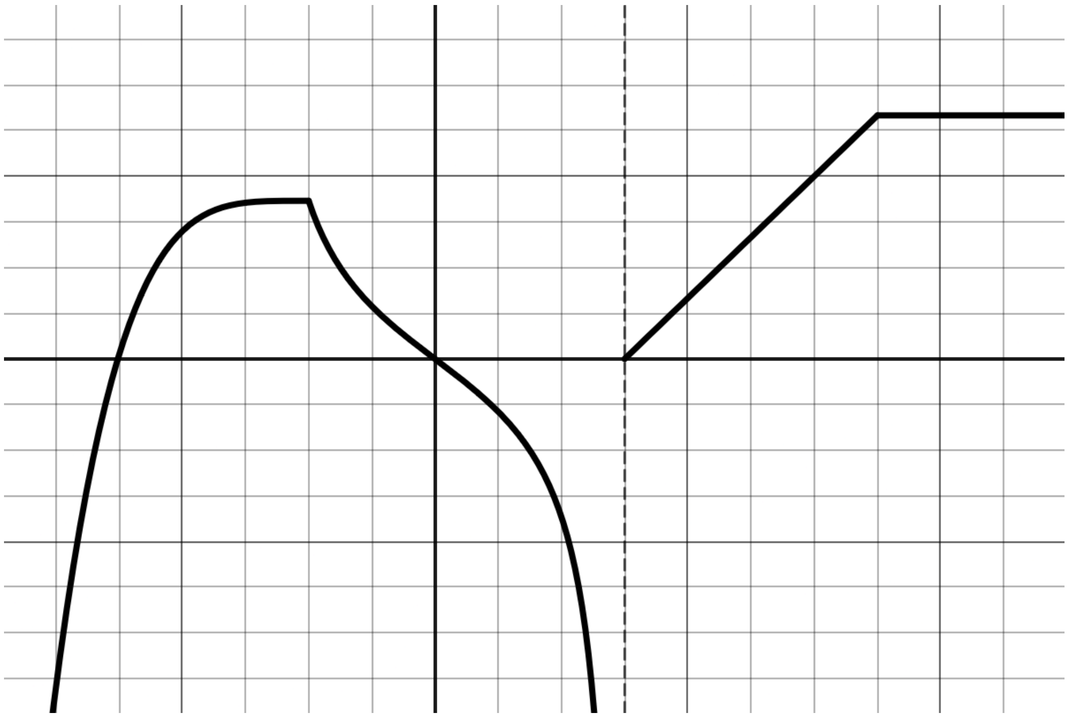


MATH 201 Your name: ____________________

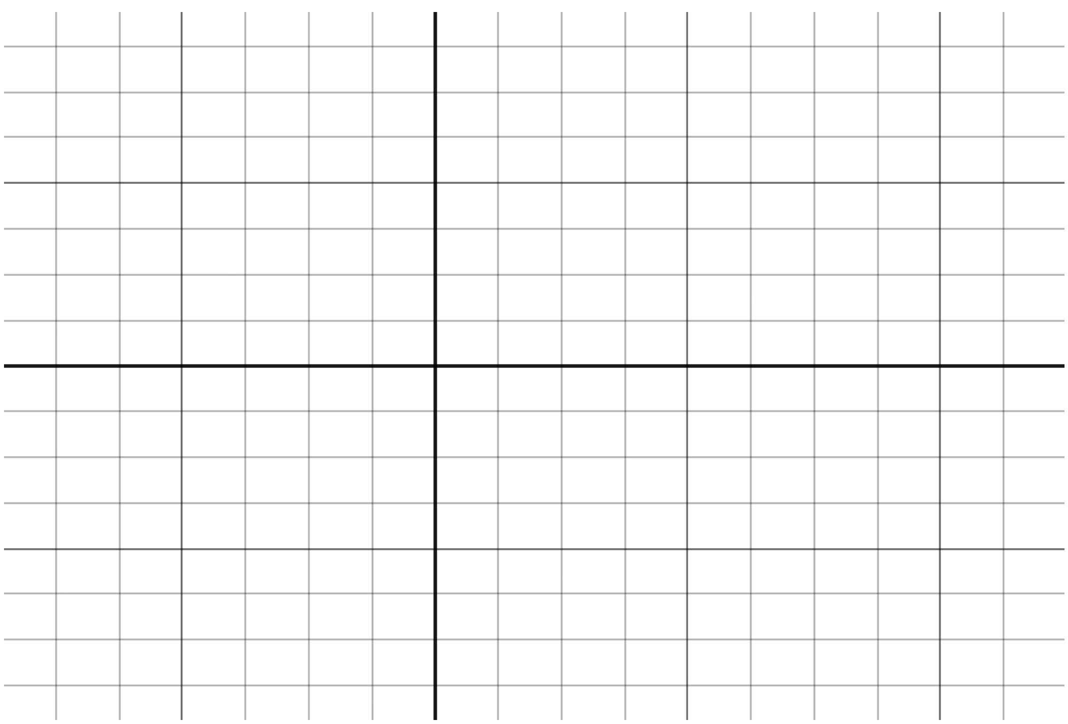
Learning target DFb, version 2

Given a graph of a function, estimate specific values of the derivative, and/or produce a sketch of the derivative function.

Here is the graph of some wacky function $h(t)$:



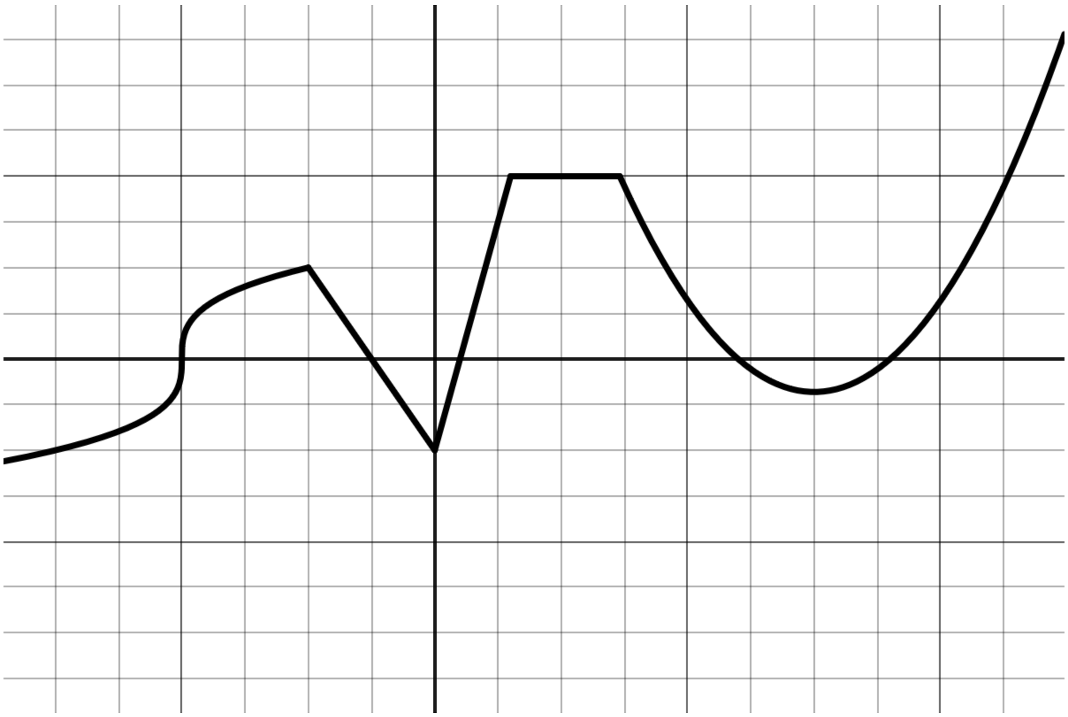
Sketch the graph of $h'(t)$ on the blank axes below.



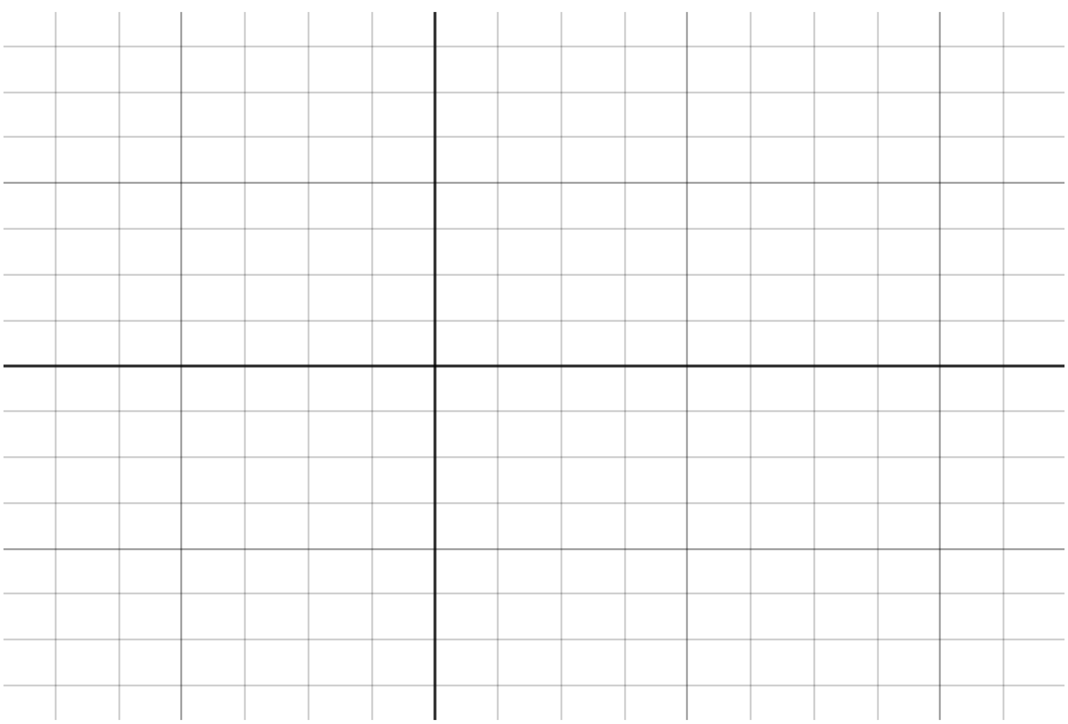
Learning target DFb, version 3

Given a graph of a function, estimate specific values of the derivative, and/or produce a sketch of the derivative function.

Here is the graph of some wacky function $z(x)$:



Sketch the graph of $z'(x)$ on the blank axes below.

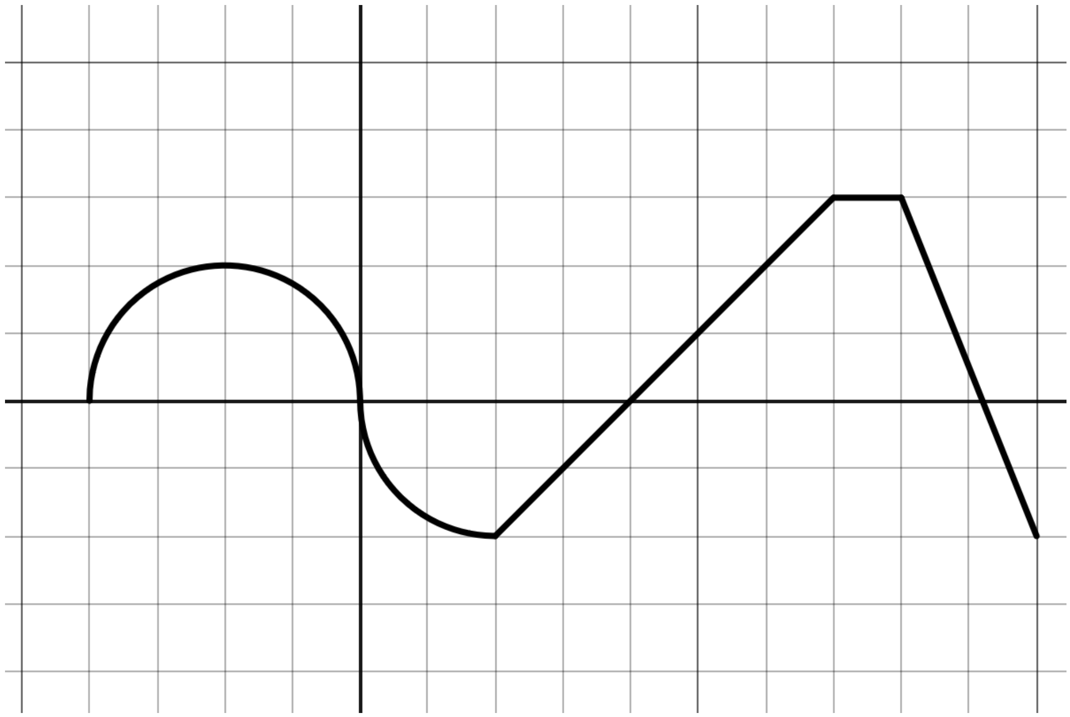


MATH 201 Your name: ____________________

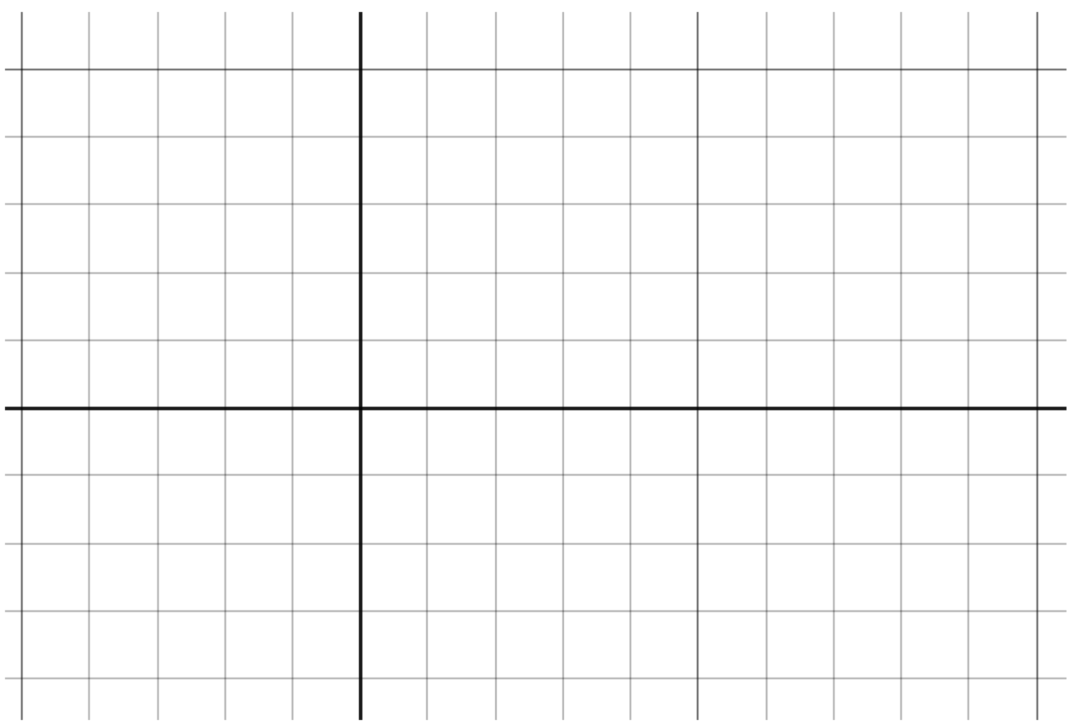
Learning target DFb, version 4

Given a graph of a function, estimate specific values of the derivative, and/or produce a sketch of the derivative function.

Here is the graph of some wacky function $g(t)$:



Sketch the graph of $g'(t)$ on the blank axes below.



MATH 201 Your name: ________________________

Learning target AD2, version 2

Suppose that for some function $p(x)$, you know the following information:

- $p(-2) = 5$,
- $p'(-2) = 1.5$,
- $p''(x) < 0$ for x-values close to -2.

1. Explain and demonstrate how to find the linearization $L(x)$ of $p(x)$ at $x = -2$.

2. Explain and demonstrate how to estimate the value of $p(-2.03)$ using this linearization.

3. Is your estimate of $p(-2.03)$ greater than or less than the actual value? How do you know?

4. Sketch a possible graph of $p(x)$ and its linearization $L(x)$ nearby $x = -2$ to illustrate your findings. Label important points in your sketch with their coordinates.

MATH 201 Your name: ____________________

Learning target AD2, version 3

The funny thing about square roots is that most of the time, they are *irrational* numbers, which means that they can't be written as fractions, and thus their decimal expansions go on forever and never repeat. For instance, $\sqrt{67} = 8.18535277187244996995370372473392945888048681549803996306671520$ That is clearly not a particularly useful answer to the question "what's $\sqrt{67}$?" We might in particular prefer to come up with a *fraction* that is a reasonably good approximation.

(a) The number $\sqrt{67}$ is an output y-value of the function $f(x) = \sqrt{x}$. What's the corresponding input x-value?

(b) Think of an x-value that's close to the input value you said in part (a), but for which it is *easy* to figure out the square root.

$f(\_\_\_\_) = \sqrt{} = \_\_\_\_$

(c) Compute $f'(x)$ and plug in your easy x-value. Leave your answer as a fraction.

$f'(\_\_\_\_) = \_\_\_\_$

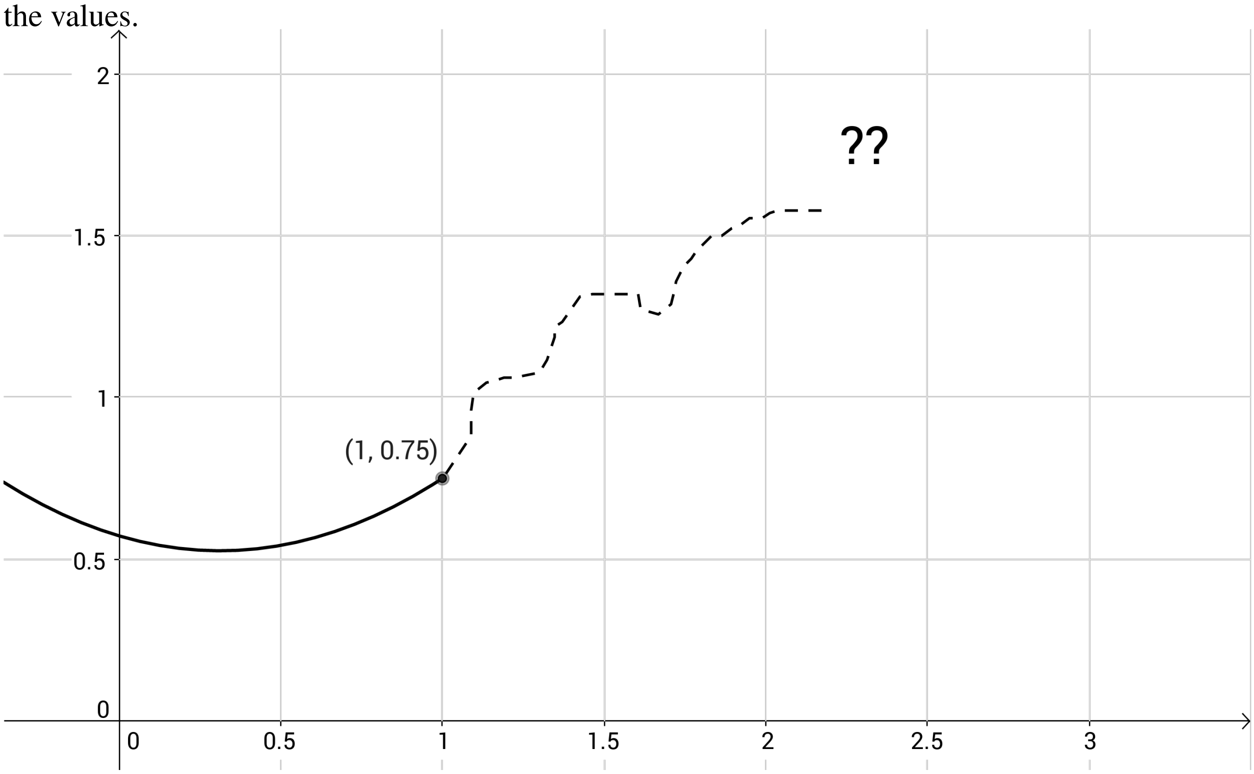
(d) Write down the equation for the tangent line to $f(x)$ at the easy x-value.

$L(x) =$

(e) Lastly but not leastly, plug in the hard x-value into the equation for the tangent line. Leave your answer as a fraction.

Learning target AD2, version 4

Part of the graph of this function $f(x)$ has been erased. We'll use linear approximation to recover the values.



(a) Draw the tangent line to this function at $x = 1$.

(b) Suppose we know that $f'(1) = 0.65$. Use point-slope form to write down an equation for $L(x)$, the tangent line to this function at $x = 1$.

(c) Use your equation for $L(x)$ to estimate $f(1.4)$.

(d) Do you think it is more likely that this is an overestimate or an underestimate? How come?

MATH 201 Your name: ______________________________

Learning target DF3, version 1

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(x) = 2\cos(x) + 4e^x$

2. $h(t) = \sqrt[3]{t^4} + \dfrac{4}{t^4}$ (Don't use the quotient rule or the chain rule.)

3. $g(w) = 6w^3 - 2w^2 - 9w - 2$

4. $C(t) = 64(e^7 + 7^e) - \dfrac{\sqrt[27]{193262}}{16 - \pi^2} + \dfrac{67}{8675309}$ (Hint: Look carefully!)

MATH 201 Your name: ______________________

Learning target DF3, version 2

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = \sqrt[4]{t^3} + \frac{7}{t^4}$ (Don't use the quotient rule or the chain rule.)

2. $f(x) = -x^5 - 4x^4 + 5x - 4$

3. $h(y) = -4\ln(y) + 3\sin(y)$

MATH 201 Your name: ______________________

Learning target DF3, version 3

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(y) = -2\,e^{y} - 4\,\sin(y)$

2. $h(x) = 9x^5 - x^4 - x^3 - 6$

3. $g(t) = \sqrt[3]{t^7} + \dfrac{2}{t^2}$ (Don't use the quotient rule or the chain rule.)

MATH 201 Your name: ____________________

Learning target DF3, version 4

Compute basic derivatives using algebraic rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = -5t^4 - 9t^3 - 6t^2 - 6$

2. $f(w) = \sqrt[3]{w^5} + \dfrac{4}{w^5}$ (Don't use the quotient rule or the chain rule.)

3. $h(y) = 2e^y + 2\sin(y)$

MATH 201 Your name: ______________________

Learning target DF4, version 1

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(x) = \frac{\ln(x)}{2x^2+6x-3}$

2. $f(x) = \left(x^2-2x-4\right)\cos(x)$

3. $h(t) = -\frac{3t^2+2t+3}{t^3}$

MATH 201 Your name: ______________________

Learning target DF4, version 2

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $h(w) = -\frac{\cos(w)}{6w^2+5w+4}$

2. $g(w) = \frac{3w^2+5w-1}{w^8}$

3. $f(w) = -\left(4w^2+6w+1\right)\ln(w)$

MATH 201 Your name: ____________________

Learning target DF4, version 3

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(w) = -\frac{2w^2 + 6w + 1}{w^{1/5}}$

2. $f(w) = -\frac{\sin(w)}{3w^2 - 3w + 4}$

3. $h(w) = -\left(2w^2 + 5w - 5\right)e^w$

MATH 201 Your name: ______________________

Learning target DF4, version 4

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $f(x) = -\left(3x^2 - 6x + 2\right)\cos(x)$

2. $h(y) = \dfrac{3y^2 - 3y - 2}{\sin(y)}$

3. $g(w) = \dfrac{2w^2 - w + 4}{w^{1/6}}$

MATH 201 Your name: ____________________

Learning target DF5, version 1

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(t) = 16\left(t + e^t + 2\right)^4$

2. $f(w) = 5\,\sin\left(w^{\frac{7}{3}}\right)$

3. $g(x) = 5\left(\sin(x)\right)^{\frac{7}{3}}$

4. $p(y) = 2\ln\left(\sin(y) + y^2 + 3\right)$

MATH 201 Your name: ______________________

Learning target DF5, version 2

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(x) = -9\cos\left(-5x^2 + 3x + 4\right)$

2. $f(y) = -9\sin\left(y^{7/2}\right)$

3. $g(t) = -9\left(\sin(t)\right)^{7/2}$

4. $k(w) = (3w + e^w - 1)^4$

MATH 201 Your name: ____________________

Learning target DF5, version 3

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $k(w) = -(5w + 2e^{w} + 1)^3$

2. $g(y) = 3\sin\left(y^{2/5}\right)$

3. $h(x) = 6\cos\left(-5x^2 - x + 4\right)$

4. $f(t) = 3\left(\sin(t)\right)^{2/5}$

MATH 201 Your name: ______________________

Learning target DF5, version 4

Compute derivatives using the Chain Rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $g(t) = 3\cos\left(-3t^2 - 3t + 4\right)$

2. $h(y) = 5\sin\left(y^{7/2}\right)$

3. $k(w) = 5\left(\sin(w)\right)^{7/2}$

4. $f(x) = -(5x + 3e^x - 4)^5$

MATH 201 Your name: ________________________

Learning target DF6, version 1

Compute derivatives of complicated functions using a combination of derivative rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $g(x) = \sqrt{\cos\left(-2x^4 + 5\right)}$

2. $f(w) = \left(\frac{6w^6 - 1}{5w + 3}\right)^4$

3. $h(y) = \left(3y^2 + 7y\right)^4 \cdot y^{1/4}$

MATH 201 Your name: ______________________

Learning target DF6, version 2

Compute derivatives of complicated functions using a combination of derivative rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $f(w) = \left(\dfrac{5w^6+1}{6w^5+2}\right)^6$

2. $g(x) = \left(5x^5 - 2x^3\right)^6 \cdot \sqrt{x}$

3. $h(y) = \sqrt{\sin\left(-2y^4+4\right)}$

MATH 201 Your name: ______________________

Learning target DF6, version 3

Compute derivatives of complicated functions using a combination of derivative rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $f(t) = \sqrt{\cos\left(-t^4 - 6\right)}$

2. $h(x) = \left(\dfrac{5x^5 - 2}{3x^3 - 1}\right)^4$

3. $g(w) = \left(3w^3 + 2w^2\right)^3 \cdot w^{2/3}$

MATH 201 Your name: ______________________________

Learning target DF6, version 4

Compute derivatives of complicated functions using a combination of derivative rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (chain, product, quotient, sum and difference, etc.) you are using.

1. $h(t) = \sqrt{\cos\left(-4t^3 - t\right)}$

2. $g(w) = \left(5w^3 + 7w^2\right)^3 \cdot \sqrt{w}$

3. $f(y) = \left(\frac{y^2 - 5}{3y^3 + 4}\right)^5$

MATH 201 Your name: ______________________

Learning target DF7, version 1

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$xy^3 + 2xy = 6$$

1. Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. Notice that the point $(2,1)$ lies on the curve $xy^3 + 2xy = 6$. Find the exact slope of the tangent line to the curve at $(2,1)$. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: ______________________________

Learning target DF7, version 2

Compute derivatives of implicitly-defined functions.

The equation $x^3 - y^3 = 9xy$ makes a curve called the *folium of Descartes*.

1. Use implicit differentiation to find an expression for $\dfrac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

2. Find the exact slope of the tangent line to this curve at the point $(-2,4)$. Show clearly how you determined this value, and label your work with proper notation.

3. (Also AD2) Find an equation for the tangent line to this curve at this point, and use your tangent line to make a reasonable guess about the y-coordinate of the point $(-2.5, \underline{\quad\quad})$.

Learning target DF7, version 3

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$x^5 - 8y^4 = \sin(y) - 7.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $(-1.4758, 0)$ (or, pretty close to that). Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: ______________________

Learning target DF7, version 4

Compute derivatives of implicitly-defined functions.

Consider the curve defined implicitly by the equation

$$\sin(x+y)+\cos(x-y)=1.$$

(a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y. Show all of your work and thinking clearly, using correct notation.

(b) This curve contains the point $\left(\frac{\pi}{2},\frac{\pi}{2}\right)$. Find the exact slope of the tangent line to this curve at this point. Show clearly how you determined this value, and label your work with proper notation.

MATH 201 Your name: ____________________

Learning target AD3, version 1

A wizard is developing a shrinking spell, but he needs to make sure it will work with his pointy hat, which is a cone whose height is three times its radius. Once he casts the spell, the height of his hat shrinks by 2 inches every second. How fast is the volume of his hat changing when the hat is just 3 inches tall? Give units on your answer.

Hints: Draw three pictures, label variables and constants, write down what you know and what you want to know, relate the variables, find the derivative wrt time, substitute and solve.

The volume of a cone is a third of the volume of the cylinder that contains it: $V = 1/3 \cdot \pi r^2 h$.

MATH 201 Your name: ____________________

Learning target AD3, version 2

Suppose an oil spill in the ocean formed a circle and was growing at a rate of 53 $\text{feet}^2/\text{minute}$. When the oil spill reaches a radius of 35 feet, how fast is the radius of the oil spill growing?

Make sure to do all six steps of the appropriate recipe.

MATH 201 Your name: ____________________

Learning target AD3, version 3

Gravel is being dumped from a conveyor belt at a rate of 12 cubic feet per minute. The gravel forms a pile in the shape of a right circular cone whose base diameter and height are always the same. How fast is the height of the pile increasing when the pile is 12 ft high?

MATH 201 Your name: ____________________

Learning target AD5 and AD4, version 1

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.
AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

(AD5)

Consider the function $f(x) = 2x^3 + 24x^2 - 54x + 7$.

1. Find the critical points using algebra, and draw a sign chart (aka labeled number line).
   Use the sign chart to decide what kind of extremum happens at each critical number.

(AD4)

2. Restricting the domain of this function to the interval $[-8,2]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-10,2]$?

MATH 201 Your name: ____________________

Learning target AD5 and AD4, version 2

AD5: Draw annotated number lines and apply Derivative Tests to classify the critical points as local extrema.
AD4: Use the Extreme Value Theorem to find the global maximum and minimum values of a continuous function on a closed interval.

(AD5)

Consider the function $g(x) = -2x^3 + 18x^2 + 42x - 100$.

1. Find the critical numbers using algebra, and draw a sign chart (aka labeled number line).
   Use the sign chart to decide what kind of extremum happens at each critical number.

(AD4)

2. Restricting the domain of this function to the interval $[-4, 4]$, what are the absolute maximum and absolute minimum values of this function, and where do they occur?

3. What if we change the domain to $[-4, 10]$?

MATH 201 Your name: ______________________________

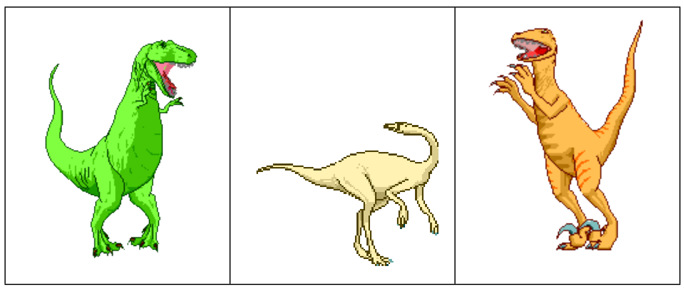
Learning target AD8, version 1

Model and analyze scenarios using applied optimization.

Hints: Draw three pictures; label variables and constants; write a constraint equation; write an objective function that just has one variable; find any endpoints; do the finding extrema recipe.

You are a dinosaur rancher, and you need to build pens for your three dinosaurs. You decide to build an electric fence around a rectangular region with total area of 25 square kilometers, and then separate this region into three equal-size rectangular pens with two more strips of electric fence. What's the minimum length of electric fencing you need to do this?

Here is one of your three pictures.



The minimum length of fence is $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$,

and the dimensions of the dinosaur pen that make this work are $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}} \times \underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$.

MATH 201 Your name: ______________________________

Learning target AD8, version 2

Model and analyze scenarios using applied optimization.

If you take a regular 8.5" x 11" sheet of paper and cut squares out of the corners, you can fold up the flaps on all four sides to make a box (without a lid). How large should you cut the squares so that the resulting box has maximum volume?

Hints: Draw three pictures; label variables and constants; write a constraint equation; write an objective function that just has one variable; find any endpoints; do the finding extrema recipe.

The maximum volume is $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$,

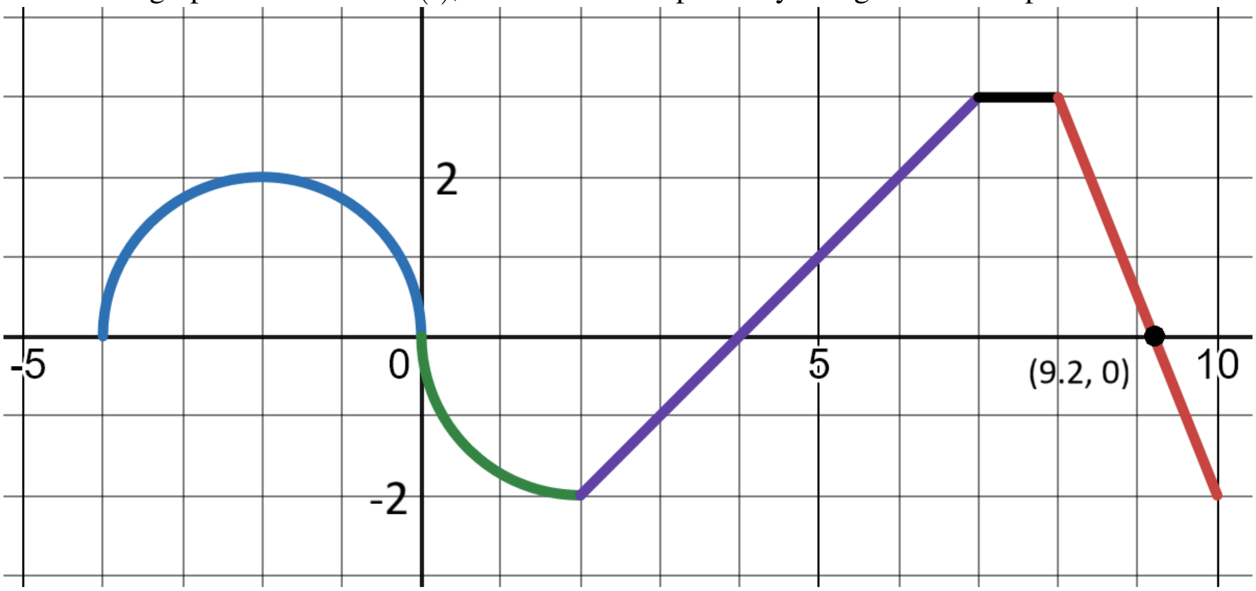
and the dimensions of the box that make this work are $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}} \times \underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}} \times \underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$.

MATH 201 Your name: ______________________________

Learning target IN1, version 1

Use geometric formulas to find definite integrals.

Here is the graph of a function $h(t)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\int_{-4}^{0} h(t)\,dt$

(e) $\int_{7}^{8} h(t)\,dt$

(b) $\int_{0}^{2} h(t)\,dt$

(f) $\int_{8}^{9.2} h(t)\,dt$

(c) $\int_{2}^{4} h(t)\,dt$

(g) $\int_{9.2}^{10} h(t)\,dt$

(d) $\int_{4}^{7} h(t)\,dt$

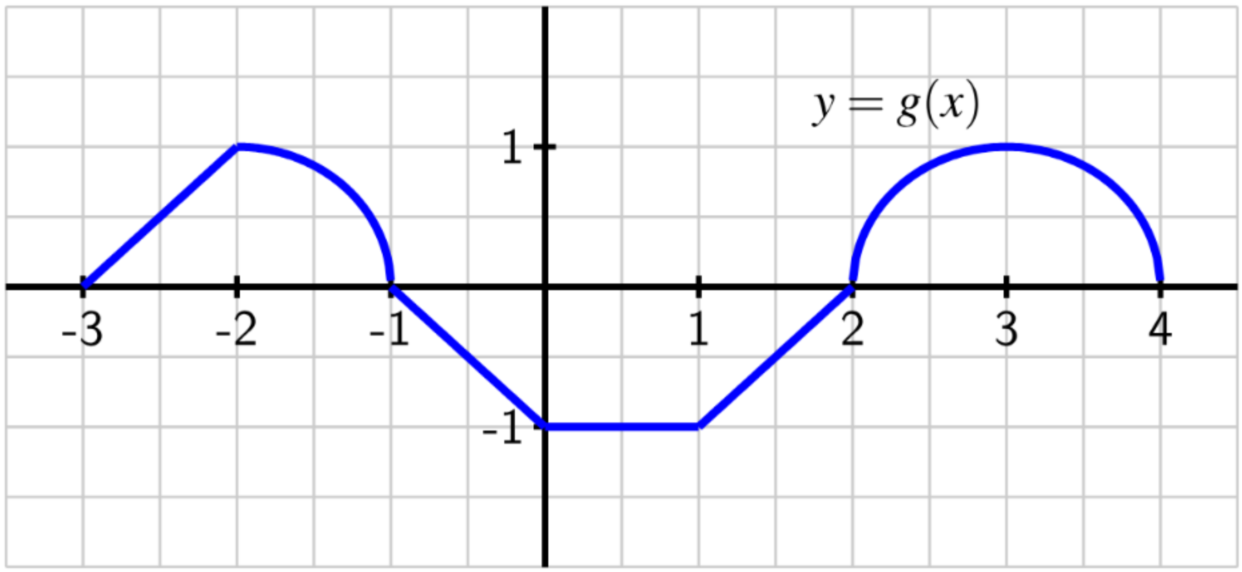
(h) $\int_{-4}^{10} h(t)\,dt$

MATH 201 Your name: ______________________________

Learning target IN1, version 2

Use geometric formulas to find definite integrals.

Here is the graph of a function $g(x)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\displaystyle\int_{-3}^{-2} g(x)\,dx$

(b) $\displaystyle\int_{-2}^{-1} g(x)\,dx$

(c) $\displaystyle\int_{-1}^{0} g(x)\,dx$

(d) $\displaystyle\int_{0}^{1} g(x)\,dx$

(e) $\displaystyle\int_{1}^{2} g(x)\,dx$

(f) $\displaystyle\int_{2}^{4} g(x)\,dx$

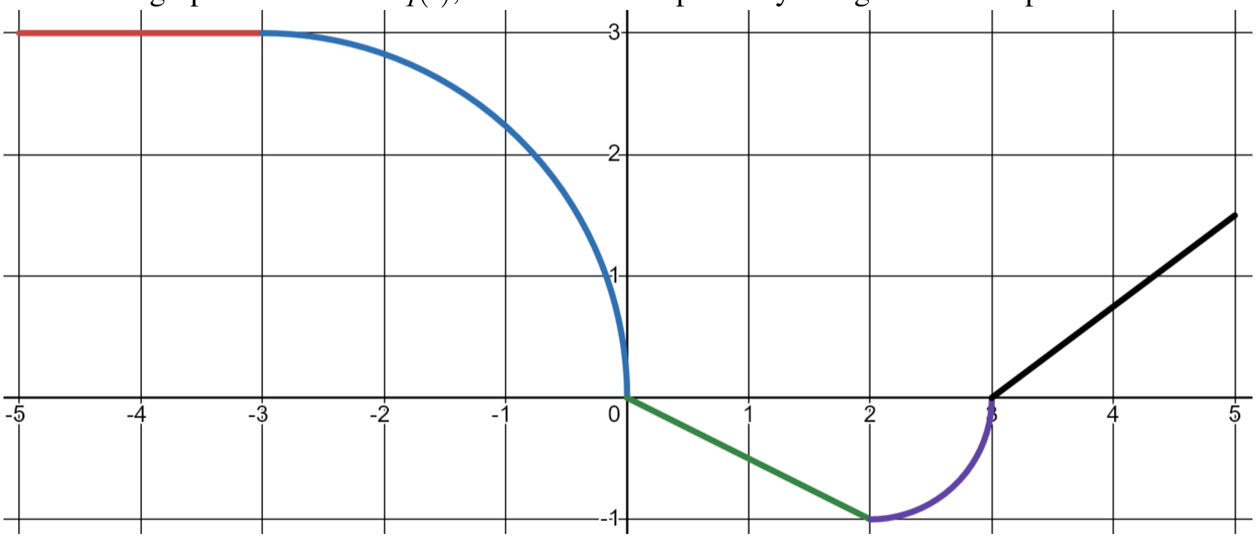
(g) $\displaystyle\int_{-3}^{4} g(x)\,dx$

MATH 201 Your name: ____________________

Learning target IN1, version 3

Use geometric formulas to find definite integrals.

Here is the graph of a function $q(t)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\int_{-5}^{-3} q(t)\, dt$

(e) $\int_{3}^{5} q(t)\, dt$

(b) $\int_{-3}^{0} q(t)\, dt$

(f) $\int_{-3}^{2} q(t)\, dt$

(c) $\int_{0}^{2} q(t)\, dt$

(g) $\int_{-5}^{5} q(t)\, dt$

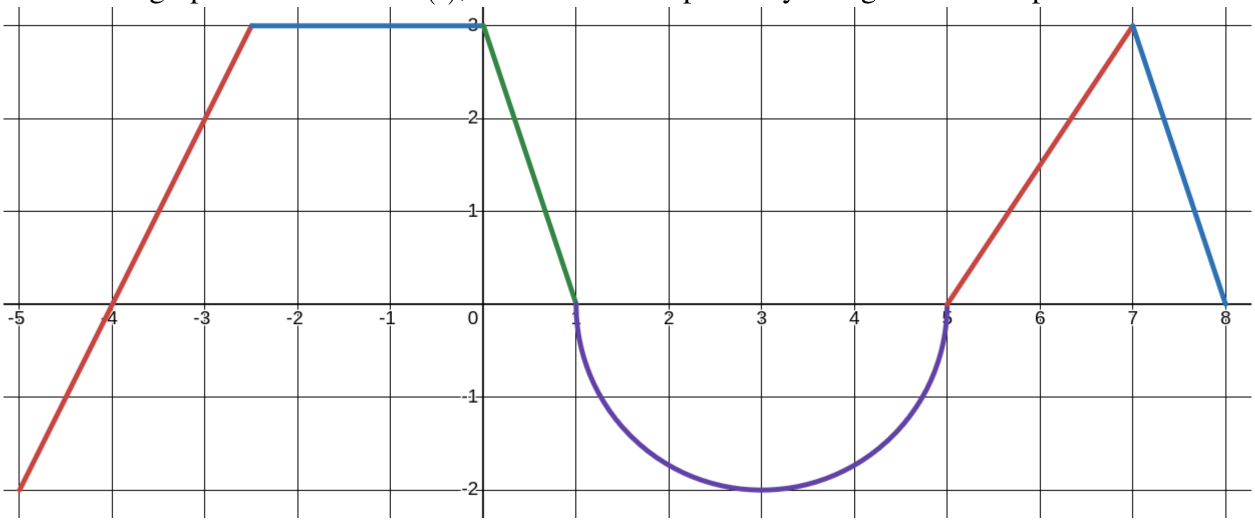
(d) $\int_{2}^{3} q(t)\, dt$

MATH 201 Your name: ______________________

Learning target IN1, version 4

Use geometric formulas to find definite integrals.

Here is the graph of a function $N(t)$, which is made up of only straight lines and parts of circles.



Compute each of the following.

(a) $\int_{-5}^{-4} N(t)\,dt$

(e) $\int_{1}^{5} N(t)\,dt$

(b) $\int_{-4}^{-2.5} N(t)\,dt$

(f) $\int_{5}^{7} N(t)\,dt$

(c) $\int_{-2.5}^{0} N(t)\,dt$

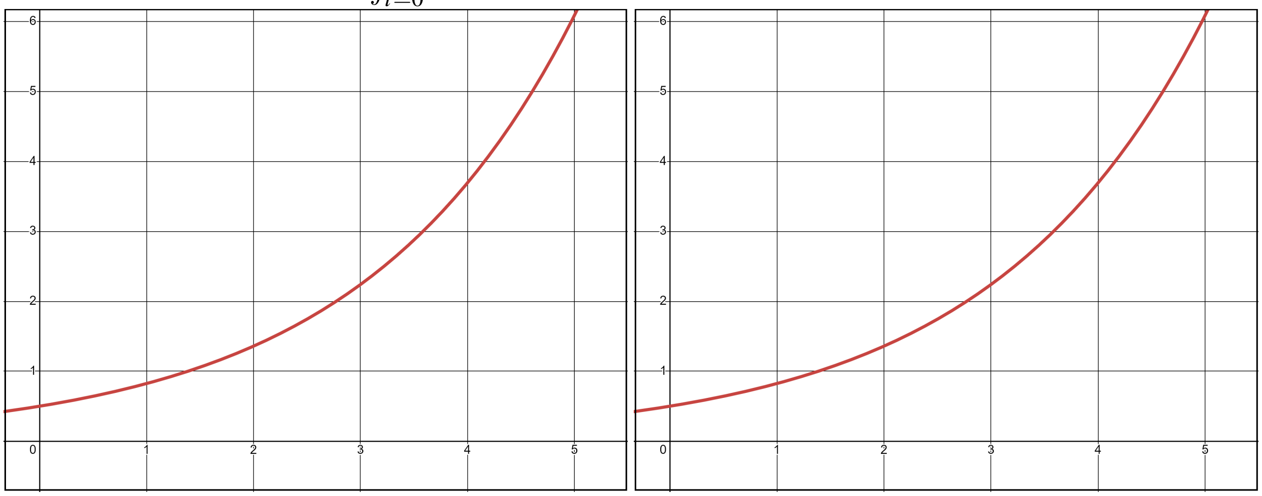
(g) $\int_{7}^{8} N(t)\,dt$

(d) $\int_{0}^{1} N(t)\,dt$

(h) $\int_{-5}^{8} N(t)\,dt$

Learning target IN2 and INx, version 1

The rate at which pollution escapes a scrubbing process at a manufacturing plant increases over time as filters and other technologies become less effective. Suppose that the rate of pollution (measured in tons per week) is given by the function r that is pictured below. Throughout this problem, we're interested in $\int_{t=0}^{5} r(t)\,dt$.



(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

(c) If $r(t) = 0.5e^{0.5t}$, compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

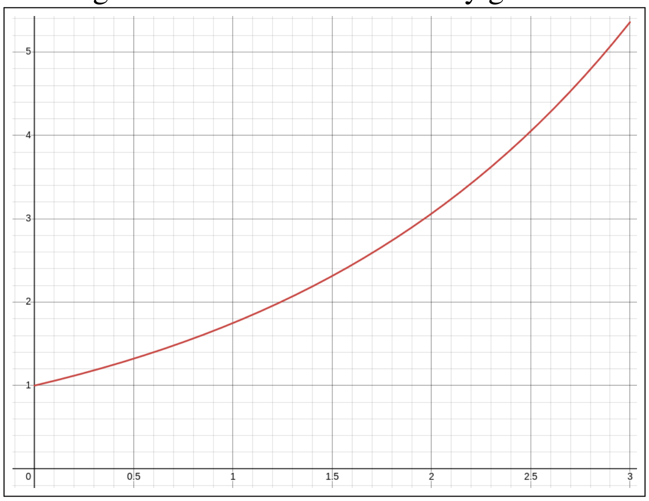
Include units on every number you write.

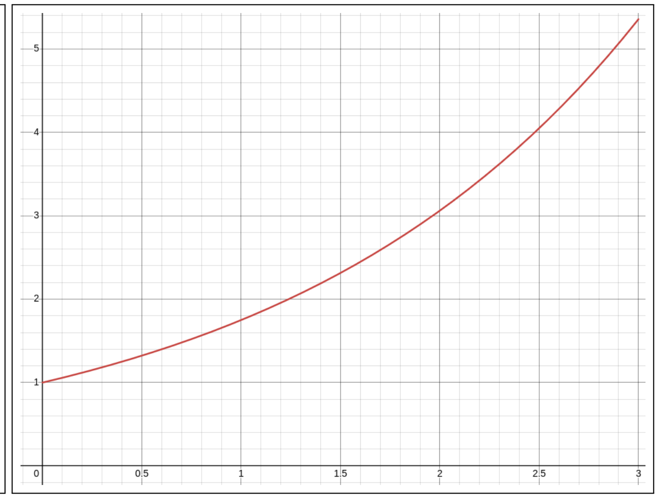
(d) Give your best guess for $\int_{t=0}^{5} r(t)\,dt$, **include units**, and explain what this number means about pollution.

MATH 201 Your name: ____________________

Learning target IN2 and INx, version 2

A colony of bacteria in a petri dish is growing at a rate of $f(t) = 1.75^t$ million bacteria per hour. Let's figure out how much the colony grows over the course of three hours.





(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

(c) Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

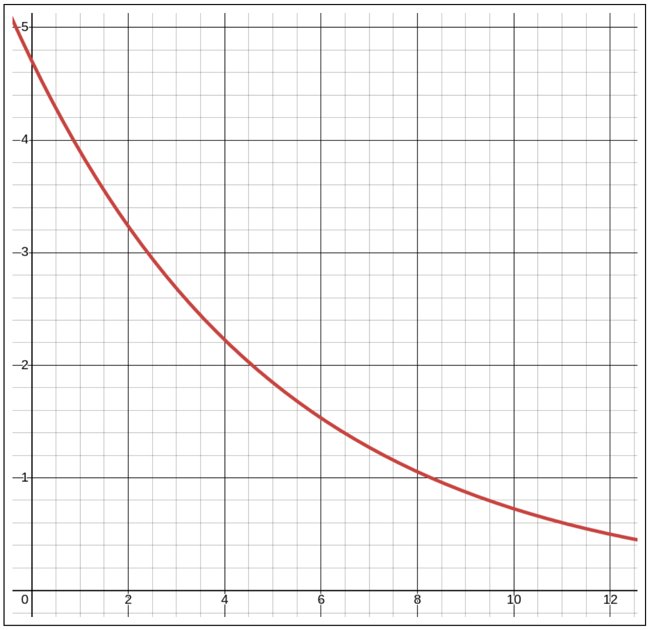
Include units on every number you write.

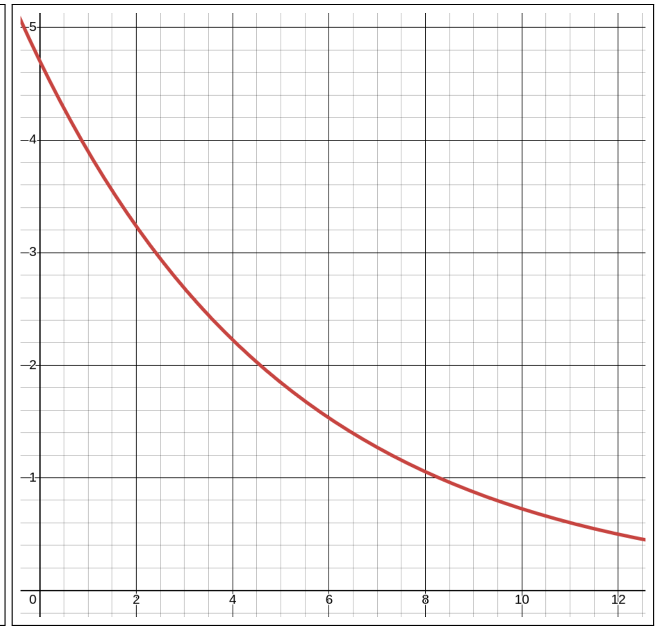
(d) Give your best guess for $\int_{t=0}^{3} f(t)\,dt$, **include units**, and explain what this number means about the bacteria colony.

Learning target IN2 and INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$





(a) On the first graph above, carefully draw the *left* Riemann sum with four rectagles of uniform width. (How wide should each rectangle be?)

(b) On the second graph above, carefully draw the *right* Riemann sum with four rectangles.

(c) Compute both of these Riemann sums. Show your work for computing the heights of the rectangles. (Hint: lots of your work will overlap!)

Include units on every number you write.

(d) Give your best guess for $\int_{t=0}^{12} p(t)\,dt$, **include units**, and explain what this number means about the puppy.

MATH 201 Your name: ______________________

Learning target INx, version 3

Over the first year of its life, a golden retriever puppy gains weight at a rate of

$$p(t) = 4.7e^{-0.187t} \text{ kilograms per month.}$$

(a) On the graph above, carefully draw a Riemann sum with four rectangles of uniform width. (How wide should each rectangle be?)

(b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the width: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$ the area: $\underbrace{\qquad\qquad}_{\text{number}}\ \underbrace{\qquad}_{\text{units}}$

(c) What would the overall area $\int_{t=0}^{12} p(t)\,dt$ mean about the puppy? **Give units in your answer.**

MATH 201 Your name: ______________________________

Learning target INx, version 4

Suppose that an underground storage tank leaks pollutants at a rate of $r(t)$ gallons per day, where t is measured in days. A researcher studying watershed quality develops the model graphed below:

$$r(t) = 0.0069t^3 - 0.14t^2 + 11.485.$$

The researcher is interested in $\int_{t=2}^{10} r(t)\, dt$.

(a) On the graph above, carefully draw a Riemann sum between $t = 2$ and $t = 10$ with four rectangles of uniform width.
(How wide should each rectangle be?)

(b) Choose one of the Riemann rectangles you drew and shade it in for clarity. For this rectangle, compute and label:

the height: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\quad}_{\text{units}}$ the width: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\quad}_{\text{units}}$ the area: $\underbrace{\qquad\quad}_{\text{number}}\ \underbrace{\quad}_{\text{units}}$

(c) What would the overall area $\int_{t=2}^{10} r(t)\, dt$ mean about the watershed? **Give units!**

Learning target IN3 and IN5, version 1

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.

Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=4}^{6} \left(\frac{6}{x}\right) dx$

(b) $\displaystyle\int_{x=\frac{5}{4}\pi}^{\frac{4}{3}\pi} \left(-2\sec^2(x)\right) dx$

(c) $\displaystyle\int_{x=-2}^{-1} \left(3x^3+10x-1\right) dx$

MATH 201 Your name: ____________________

Learning target IN3 and IN5, version 2

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.
Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=1}^{5} \left(\frac{2}{x} + 2^x\right) dx$

(b) $\displaystyle\int_{x=\pi/3}^{3\pi/4} (7\cos(x))\, dx$

(c) $\displaystyle\int_{x=-3}^{1} \left(9x^3 - 9x^2 + 2\right) dx$

Learning target IN3 and IN5, version 3

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.
Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=-2}^{4} \left(-3x^2+4x-12\right) dx$

(b) $\displaystyle\int_{x=2}^{3} \left(e^x - \frac{3}{x}\right) dx$

(c) $\displaystyle\int_{x=\pi/6}^{2\pi/3} (4\sin(x))\, dx$

MATH 201 Your name: ______________________

Learning target IN3 and IN5, version 4

Evaluate a definite integral using the Fundamental Theorem of Calculus.

Explain how to compute the exact value of each of the following definite integrals using the Fundamental Theorem of Calculus.

Leave all answers in exact form, with no decimal approximations.

(a) $\displaystyle\int_{x=0}^{\pi} \left(7\sin(x) - 2x^2\right) dx$

(b) $\displaystyle\int_{x=2}^{10} \left(\frac{3}{x} + 7^x\right) dx$

(c) $\displaystyle\int_{x=-1}^{3} \left(9x^4 - 12x^2 + 7x - 3\right) dx$